

Timely predicting Power Mosfet failure in EV: AI based methodology and deployment at the edge

Robin Faro
Research and Dev. Lab
Deepensing SRL –
Catania, Italy
robinfaro@deepensing.it

Francesco Cancelliere
Dept. of Electrical, Electronics
and Computer Engineering
University of Catania, Italy
francesco.cancelliere@phd.unict.it

Alessandro Strano
Research and Dev. Lab
Deepensing SRL –
Catania, Italy
alessandrostrano@deepensing.it

Raffaele Mineo
Dept. of Electrical, Electronics
and Computer Engineering
University of Catania, Italy
raffaele.mineo@unict.it

Abstract— Predicting the residual useful life (RUL) of power MOSFETs in electric vehicles (EVs) is crucial due to their role in transferring power from batteries to the engine. Despite its practical importance, most solutions are theoretical. This paper proposes a methodology for MOSFET RUL prediction tailored for edge devices like Nvidia Jetson ORIN and STM32. Two machine learning-based models are introduced. The first identifies the $R_{ds(on)}$ degradation law from a dataset that best matches the observed degradation, suitable for low-noise scenarios but lacking fine-tuning. The second model, using an autoencoder with dual decoders, is more robust. One decoder ensures the $R_{ds(on)}$ degradation reflects normal device behavior, while the other provides timely warnings as $R_{ds(on)}$ approaches critical limits, enabling proactive replacement. The paper validates these methods with theoretical and experimental datasets, emphasizing the second model's successful implementation on edge devices, demonstrating its practicality for EV applications.

Keywords – MOSFET failure prediction, Predictive maintenance, Electric vehicles, LSTM-based autoencoder, Edge computing.

I. INTRODUCTION

The problem of predicting the Remaining Useful Life (RUL) of power MOSFETs in electric vehicles (EVs) is increasingly studied due to their critical role in transferring power from the batteries to the engine [1]. Although a secondary MOSFET board is installed to switch in when the primary board fails, predicting MOSFET failure in advance would allow timely replacement, maintaining vehicle reliability and improving user experience. Thus, MOSFET RUL prediction remains a crucial topic in the EV industry.

Despite the practical relevance of this issue, most existing approaches are theoretical [2], [3]. Only a few studies propose practical methodologies, such as those in [4], [5]. A feasible solution should combine accurate predictions with deployment on edge devices that meet EV industry standards. For instance, edge boards must comply with a process capability (C_{pk}) of 0.33, ensuring a failure rate of less than three parts per million (ppm) [6].

This paper introduces a practical MOSFET RUL prediction methodology designed for edge devices like Nvidia Jetson ORIN and STM32. In the context of MOSFET degradation markers, $R_{ds(on)}$ (drain-source resistance), I_{ds} (drain-source current), and V_{ds} (drain-source voltage) or

their combinations are commonly cited. This study assumes $R_{ds(on)}$, in the following denoted by R_{ds} for brevity, as marker for Mosfet degradation. Critical R_{ds} threshold varies greatly, typically from 0.015 to 0.25 ohms, depending on the operating voltage, beyond which the device is near failure [7]. The goal is to alert users at least one month before R_{ds} reaches its limit to provide sufficient time for recovery and avoid underutilization of the MOSFET.

Two machine learning-based methods are proposed:

- Simple Degradation Matching: matches the observed R_{ds} degradation to the closest dataset law. Ideal for low-noise scenarios but lacks fine-tuning capabilities.
- Dual Decoder Autoencoder: a robust method using an autoencoder with two decoders. One ensures the R_{ds} degradation is normal, while the other issues timely warnings as R_{ds} nears critical levels, enabling proactive replacement.

Section 2 outlines the two proposed methods and their validation using theoretical and experimental datasets. Section 3 focuses on the fine-tuning of the second method, demonstrating its implementation on a Jetson board and discussing its feasibility on STM32. Section 4 summarizes the advantages of the proposed approach, highlights open problems, and suggests directions for future work.

II. TWO METHODS FOR MOSFET RUL PREDICTION

As motivated in the introduction, the chosen marker for identifying when the MOSFET device approaches failure is R_{ds} . Theoretically, R_{ds} follows an exponential increase over time. The formula used in this paper is inspired by [2]:

$$R_{ds(on)}(t, n) = 0.02 + \frac{2.67}{1000} \left(\exp \left(\frac{(1.15 + 0.1n) \cdot t}{100} \right) - 1 \right) \quad (1)$$

where n represents the number of months of the MOSFET total useful life (TUL), 0.02Ω is the initial resistance R_0 , whereas the term $(1.15 + 0.1 n)$ represents the degradation rate factor of the n -th device. In other terms, the MOSFET TUL is the time interval from the time at which the MOSFET starts to work whose R_{ds} is equal to R_0 , and the one at which R_{ds} reaches its maximum value. In fig.1, we illustrate the degradation curves of 36 MOSFET devices, with their TUL ranging from 24 to 60 months.

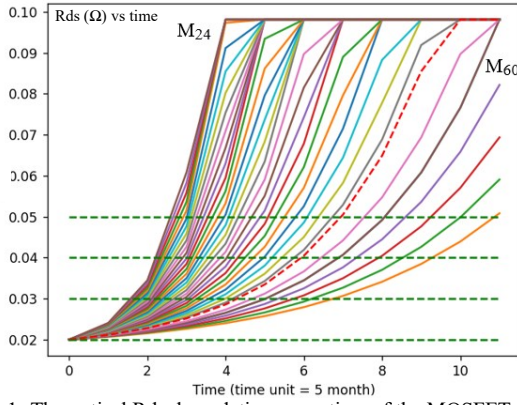


Fig. 1- Theoretical Rds degradations over time of the MOSFET devices

In fig.1 we assume, without loss of generality, that the MOSFET lifespan is five years and failures before two years are highly improbable. Also we assume, as in [8], that the maximum Rds value of the Mosfet under test (M_{ut}) is 0.05 Ω . These curves form the dataset used by our methods to estimate the RUL at time t . This estimation is based on the previous Rds values received from M_{ut} . Each device is identified by its TUL, with the worst-case device labeled as M_{24} and the best-case device as M_{60} .

A A simple method to forecast future vales of Rds in an environment few affected by noise

The first method is able to predict the unknown degradation law of M_{ut} using the above theoretical dataset. In particular the method will find the degradation law of the MOSFET, let say M_{bf} , in the dataset that best fits the degradation law of M_{ut} . Under this hypothesis, the Rds value of the real device expected after $k \Delta T$ time units is given by $Rds(t + k\Delta T, M_{bf})$.

To find M_{bf} we used an LSTM neural network able to check if an unknown device such as M_{ut} may be considered the subsequent device of M_n in the mentioned ordered sequence [9]. If not, the neural network should provide means to find the device number between M_n and M_{ut} .

For this reason, for a given n and a prefixed interval ΔR of Rds, let say from Rds_1 to Rds_2 , we should pass to the mentioned LSTM network, let say $LSTM_n$, the Rds data coming from M_{ut} in the same interval ΔR so that $LSTM_n$ may give at its output the distance in time units between the time $t_{n-1,2}$ and $t_{ut,2}$ being the former time value the time at which the subsequent device of M_n reached its maximum in the interval ΔR and the latter one the time at which M_{ut} reached its maximum in the same interval ΔR . The knowledge of $t_{n-1,2}$ and $t_{ut,2}$ allows us to find the number δ of devices in the dataset between $t_{n-1,2}$ and $t_{ut,2}$ and to find the device of the dataset that best fits the one under test given by $M_{bf} = M_{n-1+\delta}$. Therefore, the predicted value after $k \Delta T$ time units (where Δt is the time interval between $t_{n-1,2}$ and $t_{ut,2}$) is given by $Rds(t + k \Delta T, M_{n-1+\delta})$.

Since the index n may be fixed arbitrarily, we select in our method $n = 24$ (i.e., the one related to the worst device) so that we may use only one model of the LSTM network, let say $LSTM_{24}$. Thus the Rds that best fits the Rds of M_{ut} is $Rds(t + k \Delta T, M_{23+\delta})$

To predict Rds effectively, we should forecast the time when it approaches its failure value of 0.05 ohms, rather than after it reaches this threshold. This requires a step-by-step procedure, where the degradation law is identified by applying the prediction algorithm iteratively over m steps. In step m , Rds data are collected, and the Rds formula to be used for step $m+1$ is predicted. For example, if $m = 3$, the prediction process occurs over three intervals: 0.02–0.03, 0.03–0.04, and 0.04–0.05, as shown in Fig. 1. The first step predicting Rds in the interval 0.03–0.04, then the second step predicting Rds in the interval 0.4–0.05, and finally the third step verifying the prediction across the previous steps.

In principle ARIMA or other LSTM based methods can be used to analyze trends to predict Rds [10], [11]. ARIMA is less accurate [12] and LSTM offers improved precision [13], but also it may propagate errors in long-term forecasting. The proposed LSTM-based method mitigate this by using long-term predictions in which each steps does not predice the next value of Rds but the Rds curve for the next step. For low noise levels (1–2%), median interpolation of the collected Rds data can suffice to maintain a high accuracy, but for higher noise, typical in EVs, a more robust approach is required. Also, fine-tuning the method remains challenging due to limitations in maintaining dataset sequence order.

B AI based method to forecast future vales of Rds in an environment affected by noise

To overcome the limitations of the first method, we propose a second method utilizing an autoencoder with two decoders: one reconstructs Rds from M_{ut} , and the other predicts Rds. In the first method, future Rds values up to 0.05 ohms are known after the initial step, with refinement achieved by recalculating at each step the distance and number of devices from M_{24} to M_{ut} .

In the second method, the M_{ut} signal from the EV board allows its decoder to reconstruct Rds up to the current time t . This is important to signal immediately faulted Mosfets, i.e., the ones the autoencoder in not able to reconstruct by its decoder. But, another decoder is needed to predict Rds by back extending the curve related to the data coming from M_{ut} of $t+\delta T$, possibly with a higher profile so that at each time t we have both the the current value (from the lower curve) and the predict one at time $t + \delta T$ (from the higher curve).

Fig.2 illustrates this, showing the curve to be passed to first decoder for reconstruction (blue) and the prediction curve (red) to be used for training the second decoder.

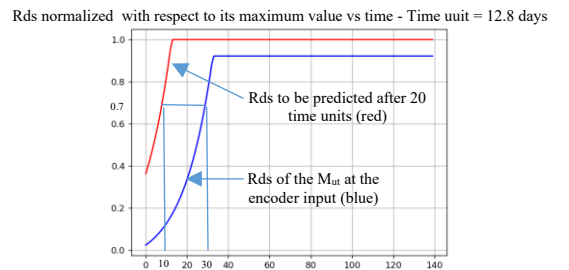


Fig.2 – Theoretical curves dealing with Rds coming from M_{ut} and the value predicted after 20 time units, e.g., at $t = 10$ we may know that the Rds value at $t + 20$ time units will be 70% of its maximum value.

This setup enables the second decoder to predict Rds in the interval from t_1 to $t_1 + \delta T$ from the prediction obtained in the first interval from t_0 to t_1 , as pointed out in fig.3, by appending

these predicted values to those received up to t_1 , the process iteratively predicts future values up to the point where Rds reaches 0.05Ω .

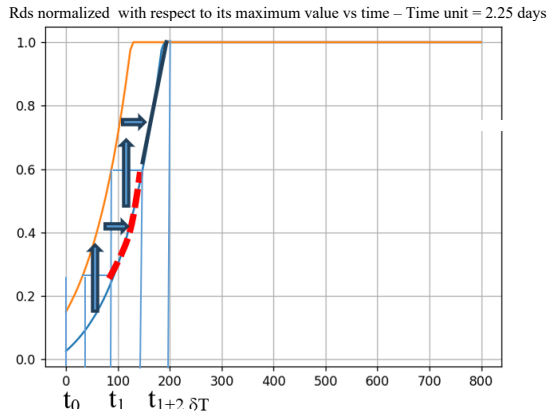


Fig.3 – Step by step procedure to predict Rds at a given step using the prediction generated by the previous step.

To assess the accuracy of the second method in both training and testing phases, we assume that the accuracy of reconstructing the current signal and its translated version by δT , using the full signal from t_0 to t_i , is close to that obtained by the step-by-step procedure extended from t_1 to t_i . This is supported by fig.4, where the forecast, using $\delta T = 20$, closely matches the real Rds values within the relevant range (0.02–0.05 ohms). The Mean Absolute Error (MAE)-based accuracy is 93% in training and 87% in testing, calculated using: $\text{Accuracy} = 1 - \text{MAE}/R$ [14]. The step-by-step procedure introduces approximations, but these do not propagate since the predicted signal is replaced with real values after δT . The approximation remains constant and approaches zero for very small δT , though reducing δT excessively shortens the prediction time, limiting effectiveness.

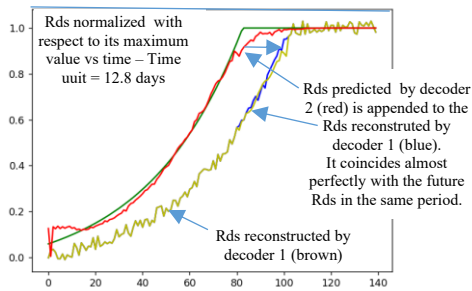


Fig.4 Prediction and reconstruction obtained by the proposed step by step procedure.

Fig. 4 allows also to point out that with a TUL of 5 years, and using 140 sampling point, the sampling period is 12,8 days (being 60 months/140=12.8 days). This choice makes $\delta T = 20$ equivalent to about 8 months, providing the driver with an alert 8 months before failure. However, an alert one month before failure is sufficient for recovery and avoids underutilization of the MOSFET. Thus, we recommend to implement two forecasts: one with $\delta T = 6$ for a yellow alert (2 months before failure) and the other with $\delta T = 3$ for a red alert (1 month before failure). For improved precision, a sampling period of $60/800 = 2.25$ days is suggested. This allows us to issue caution yellow and red alerts respectively at $\delta T = 60$ and $\delta T = 30$ (4 and 2 months before failure). Fig.5 demonstrates that the method is robust to noise, as adding 5% noise produces a forecast very close to the noise-free signal.

Rds normalized with respect to its maximum value vs time - Time unit = 2.25 days

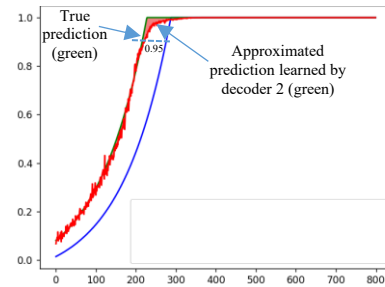


Fig.5 – Difference between the true result (green) and the approximation (red) is very low if this is made at 95% of the value at which Rds achieves its maximum value.

III. VALIDATING THE ENCODER BASED METHOD USING EXPERIMENTAL DATA AND DEPLOYING AT THE EDGE

Our primary goal is to develop MOSFET RUL prediction methods suitable for EVs, where signals are subject to noise should be fine-tuned with experimental data to enhance prediction accuracy. This implies that the first method cannot be used, Therefore, this section focuses on: a) validating the second method using experimental data from 100V power MOSFETs available on the NASA portal [15], with Rds as a failure precursor, and b) demonstrating the feasibility of deploying this method on edge devices.

A Validation of the method in the experimental setting

A preprocessing script was developed to extract Rds data from NASA portal files. As shown in fig.6, the records slightly differ from the initial dataset and are significantly noisy, making the first method unsuitable. The second method, being noise-resistant, was fine-tuned by updating autoencoder weights obtained for datasets with 800 sampling points to learn stepwise Rds degradation laws. Fig.7 shows that the experimental data used in fine tuning are the Rds signals and its shift back in time by a certain number of time units set by the user.

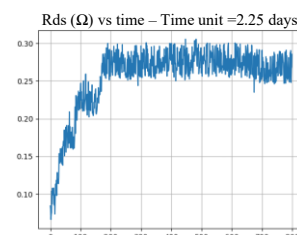


Fig.6 - One of the Rds accelerated aging experiments from the NASA dataset

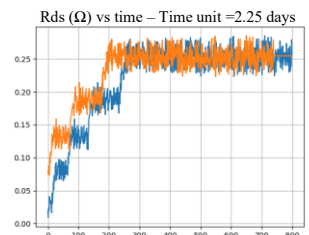


Fig.7 - Signals passed to decoders for reconstruction (blue) and prediction (red)

Fig.8 left indicates that both decoder1 and decoder2 learn satisfactory the form of the degradation laws of the experimental results.

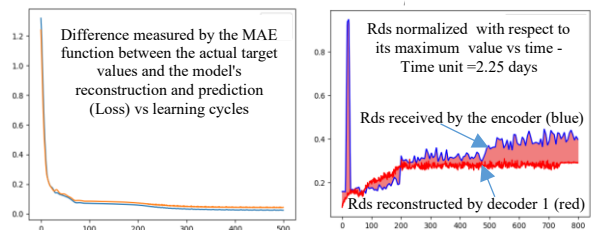


Fig.8 Training loss for both the decoders is very low (left), Anomalous MOSFETs are eliminated from the dataset (right).

After fine-tuning the autoencoder, we tested experimental data not used during training, following the procedure outlined in the previous section. Anomalous data were discarded (see fig.8 right), while the reconstruction and the prediction curves for normal data were generated by the autoencoder with high accuracy (see fig.9a and fig.9b). This accuracy is more evident if comparing the generated curves to the interpolations of the original signal and its back-translated form. Fig.9c shows that the stepwise method described in section II.B works even in this case.

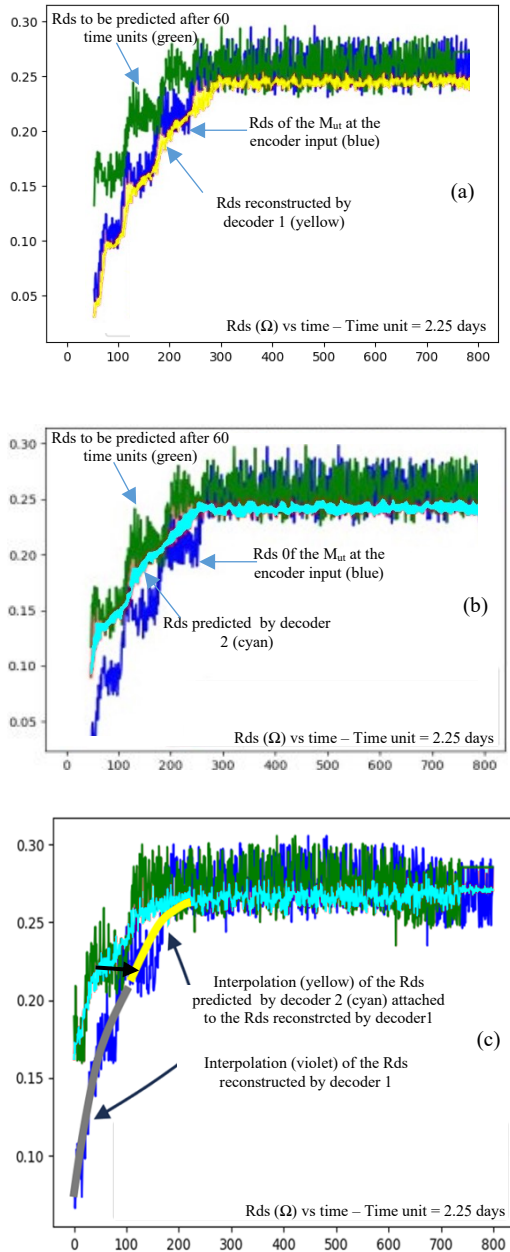


Fig.9 – a) Reconstruction of the input by decoder 1 (11 ms), indicating that the M_{ut} is not faulted. b) Prediction by decoder 2 (24 ms) with high accuracy. c) Prediction and reconstruction using the proposed step-by-step procedure, where the yellow curve represents the R_{ds} prediction. This curve is obtained by appending the corresponding segment of the prediction curve to the reconstructed curve. It aligns almost perfectly with the future real R_{ds} over the same time period.

B Deployment of the RUL prediction model at the edge

This section demonstrates that the prediction method can be implemented on edge devices with comparable accuracy and inference time. Various devices can host the MOSFET RUL prediction model, such as NVIDIA Jetson Orin, Hailo, Raspberry Pi4, and STM32. Here, we evaluate the performance of deploying the model on Jetson Orin AGX, Jetson Nano, and STM32 boards. These devices are particularly suitable as specialized versions are expected to meet EV standards, given the manufacturers' active involvement in EV applications like autonomous driving and battery/inverter control.

The MOSFET RUL prediction model proposed in Section II.B was implemented on NVIDIA Jetson ORIN AGX, Jetson ORIN Nano, and virtual on STM32 boards via the ST cloud development environment. Previous results were obtained using a Keras autoencoder on an ASUS ROG with an AMD Ryzen™ 7 4800H processor (8 cores/16 threads, 12 MB cache, 4.2 GHz max boost), while the edge device implementation was developed in PyTorch.

The Keras autoencoder achieved an inference time of 4 ms and an accuracy of 87%. On edge devices, no model quantization was required, maintaining the same accuracy across implementations. Inference times on NVIDIA systems were an order of magnitude faster than on the ASUS ROG, while STM32 performance was comparable. None of the implementations encountered memory constraints.

These results indicate that the model can be effectively deployed on NVIDIA and STM boards, provided these devices meet EV safety standards.

IV. CONCLUDING REMARKS

The paper proposed two methods for MOSFET RUL prediction. The positive results obtained in terms of accuracy and time inference encourage us to continue the study to further optimize the model in the laboratory, refining it with experimental data from accelerated aging of MOSFETs used specifically in the electric vehicles of interest.

Further evaluation is planned by mounting a MOSFET control board on cars on the track. A more detailed analysis with what is proposed by other AI-based methods using the NASA dataset is also in progress.

Evaluation of the feasibility of implementing the RUL prediction model obtained by the second method on edge devices such as Hailo and Raspberry Pi4 is planned for future work. Additionally, the use of simple Google cards or similar devices for implementing the first method will be investigated to be used in low noise scenarios.

ACKNOWLEDGMENTS

The paper was produced with the support of the Neurokit2E project (No. 101112268), funded by the EC Horizon Program (HORIZON-KDT-JU-2022-2-RIA-Topic-1) .

REFERENCES

- [1] B. Shi *et al.*, A review of silicon carbide MOSFETs in electrified vehicles: Application, challenges, and future development, *IET Power Electronics*, 17 May 2023
- [2] J. Ni, Z. Chenguang, X. Zhang and T. Lei, Remaining Useful Life Prediction Method for MOSFET Based on Time Series Model, IEEE Asia-Pacific Conference on Image Processing, Electronics and Computers (IPEC), 2021
- [3] Q. Wu, B. Xu, L. Xiao and Q. Wang, A remaining useful life prediction method of SiC MOSFET considering failure threshold uncertainty, *IET Power Electronics*, 20 November 2023
- [4] F. Rundo *et al.*, Intelligent Electrical Assessment of Silicon and Silicon Carbide Wafers for Power Applications in Automotive Field, IEEE International Workshop on Metrology for Automotive (MetroAutomotive), Bologna, Italy, 2024, pp. 18-23, doi: 10.1109/MetroAutomotive61329. 2024
- [5] C. Pino *et al.*, "Intelligent Traction Inverter in Next Generation Electric Vehicles: The Health Monitoring of Silicon-Carbide Power Modules," in *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 12, Dec. 2023
- [6] A. Kampker, K. Kreisköther, M. K. Büning, P. Treichel and J. Theelen, "Automotive quality requirements and process capability in the production of electric motors, 7th International Electric Drives Production Conference (EDPC), Würzburg, Germany, 2017
- [7] W. Xiangyu *et al.*, A data-driven lifetime prediction method for thermal stress fatigue failure of power MOSFETs." *Energy Reports*, 2022
- [8] I. C. Guran, A. Florescu, L. A. Perișoara, A Novel ON-State Resistance Modeling Technique for MOSFET Power Switches, *Mathematics* 11, 72, 2023
- [9] N. Hebbar, Time series forecasting, https://github.com/nachi-hebbar/Time-Series-Forecasting-LSTM/blob/main/RNN_Youtube.ipynb
- [10] M. S. Ibrahim *et al.*, Long-Term Lifetime Prediction of Power MOSFET Devices Based on LSTM and GRU Algorithms, *Mathematics* 11(15), 2023
- [11] C. S. Kulkarni, J. R. Celaya, G. Biswas and K. Goebel, Prognostics of Power Electronics, methods and validation experiments, 2012 IEEE AUTOTESTCON Proceedings, Anaheim, CA, USA, 2012
- [12] S.S. Namini *et al.*, A Comparison of ARIMA and LSTM in Forecasting Time Series, Conference: 17th IEEE International Conference on Machine Learning and Applications (ICMLA), 2018
- [13] T. Xu, D. Pi and S. Zeng, Remaining useful life prediction based on spatiotemporal autoencoder. *Multimed Tools Appl* 83, 2024
- [14] R. J. Hyndman and A. B. Koehler, Another look at measures of forecast accuracy, *International Journal of Forecasting*, Volume 22, Issue 4, 2006
- [15] J. R. Celaya *et al.*, Prognostics Approach for Power MOSFET under Thermal-Stress Aging Open Article NASA Portal 2011